

# Threaded Fasteners



# TOPICS

- Introduction
- Thread terminology
- Thread drawing and dimensioning
- Threaded fastener



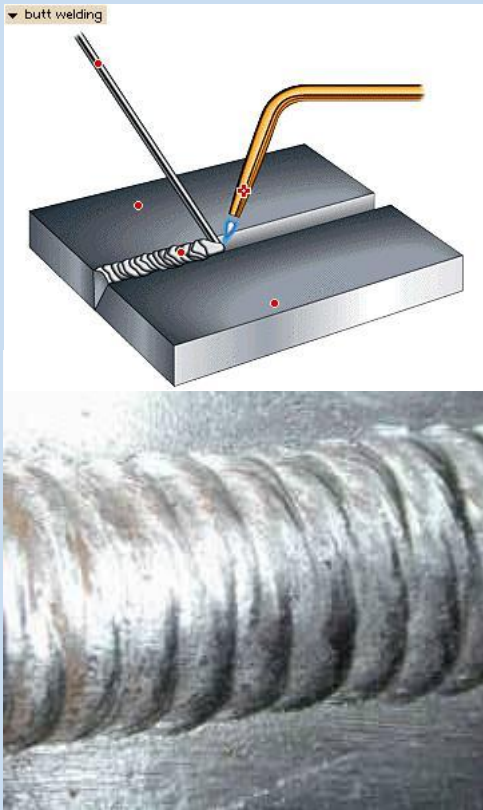
# *Introduction*



# FASTENING TYPE

## 1. Permanent

Welding



Gluing



Riveting



# FASTENING TYPE

## 2. Temporary

### 2.1 *Threaded fastener*

- bolts
- studs
- screws



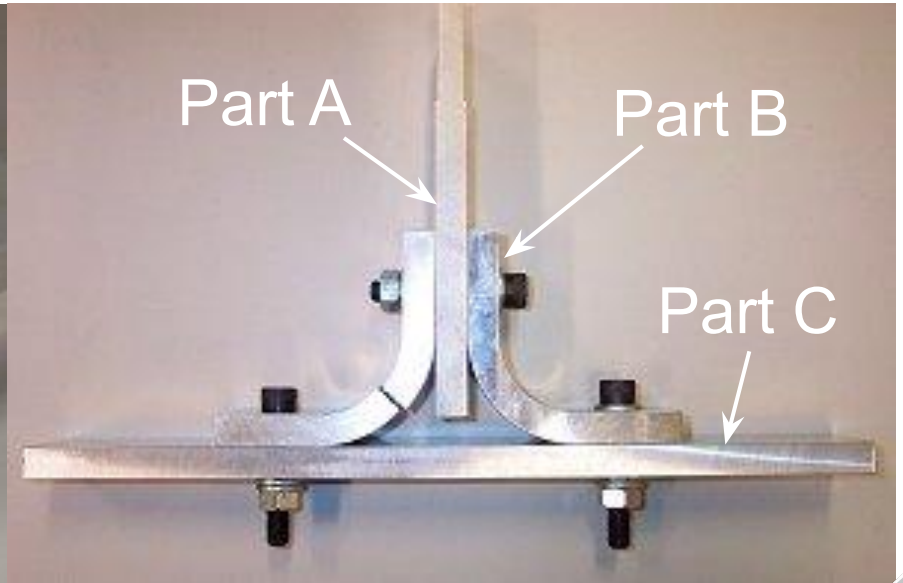
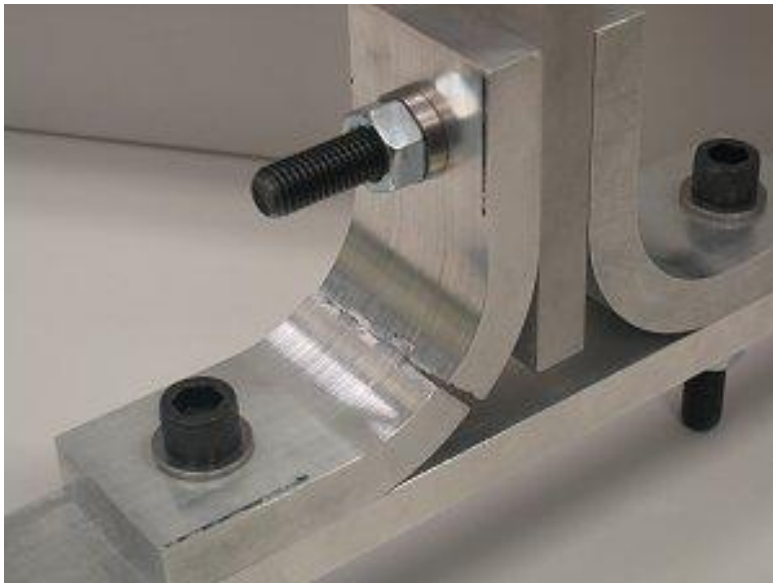
### 2.2 *Non-threaded fastener*

- keys
- pin



# THREAD APPLICATION

1. To hold parts together.
2. To move part(s) relative to others.



# THREAD APPLICATION

1. To hold parts together.
2. To move part(s) relative to others.



Wood working vise



Palm fruit pressing machine



# *Thread Terminology*





# THREAD TERMINOLOGY

***External (male)  
thread***

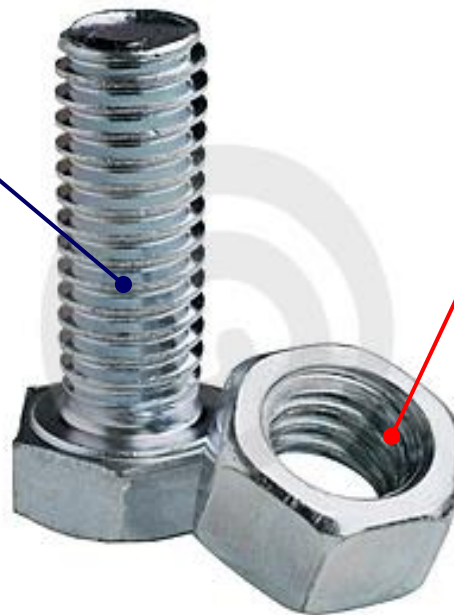
A thread cut on the **outside** of a cylindrical body.

***Internal (female)  
thread***

A thread cut on the **inside** of a cylindrical body.

***External thread***

***Internal thread***



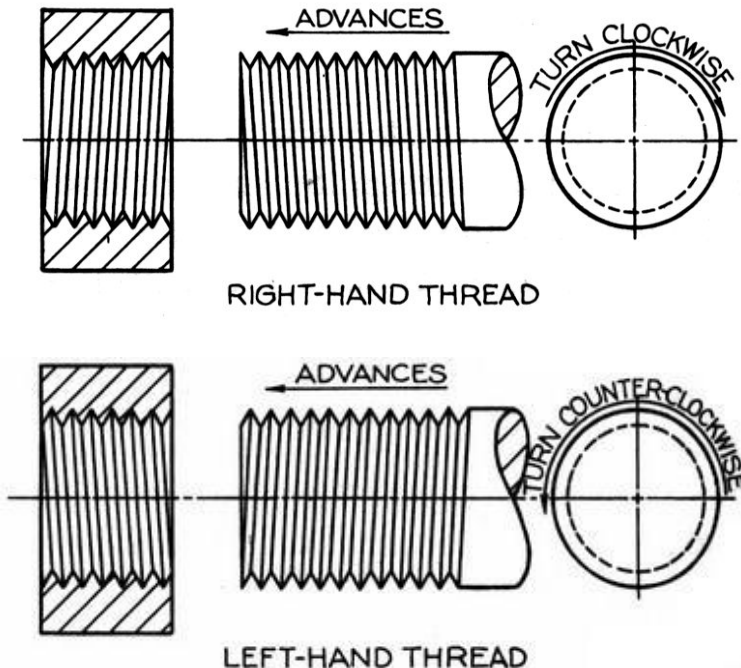
# THREAD TERMINOLOGY

**Right-hand  
thread**

Thread that will *assemble* when turned *clockwise*.

**Left-hand  
thread**

Thread that will *assemble* when turned *counter-clockwise*.



**Turnbuckle** use RH and LH thread at each end to double displacement.



# THREAD TERMINOLOGY

## ***Crest***

The ***peak edge*** of a thread.

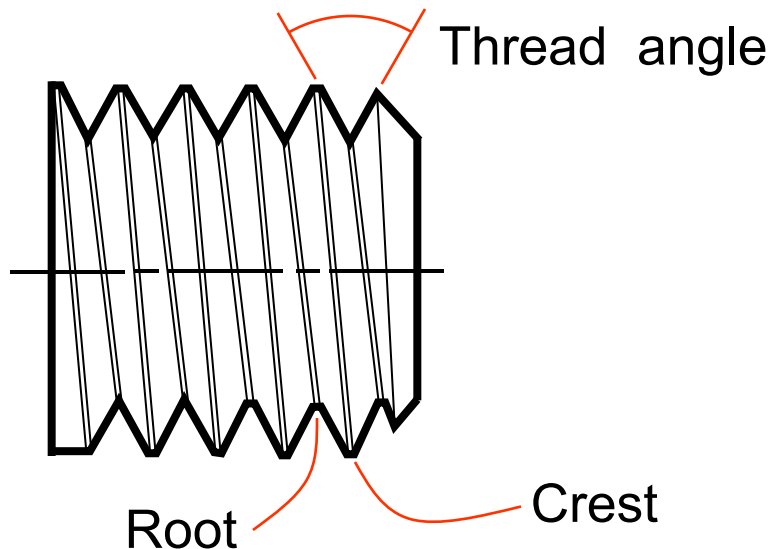
## ***Root***

The ***bottom*** of the thread cut into a cylindrical body.

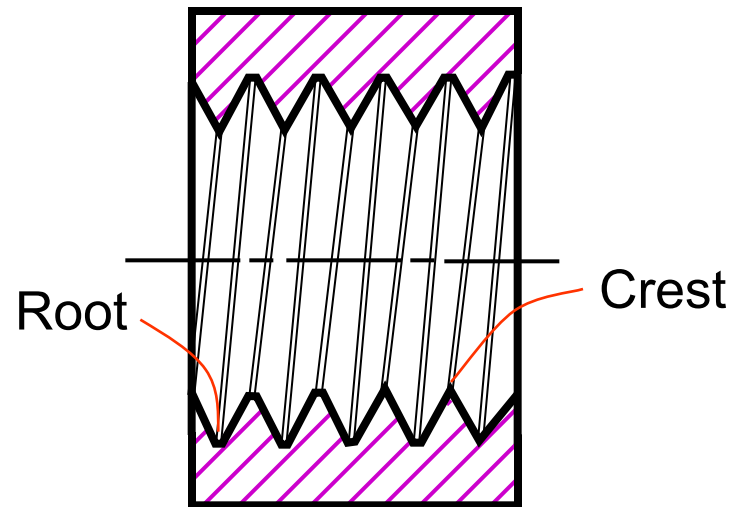
## **Thread angle**

The angle between threads faces.

### **External Thread**



### **Internal Thread**



# THREAD TERMINOLOGY

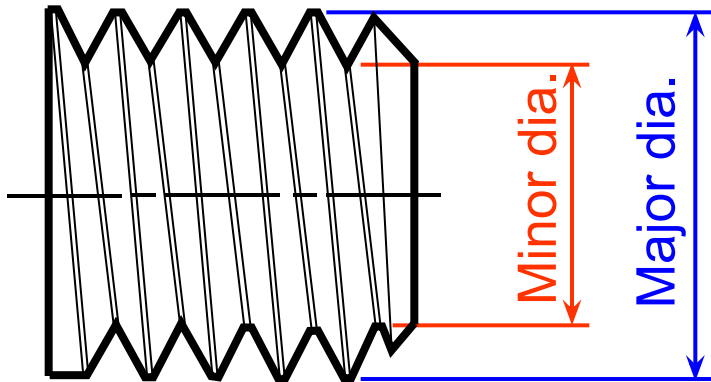
## ***Major diameter***

The ***largest diameter*** on an internal or external thread.

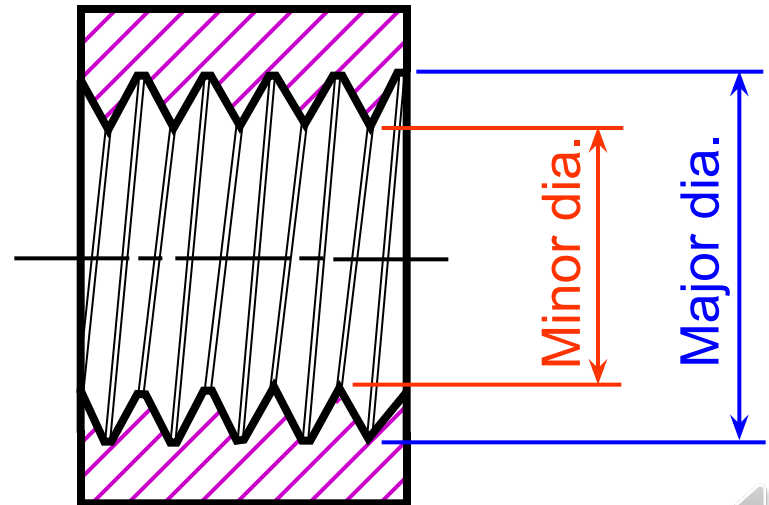
## ***Minor diameter***

The ***smallest diameter*** on an internal or external thread.

### **External Thread**



### **Internal Thread**



# THREAD TERMINOLOGY

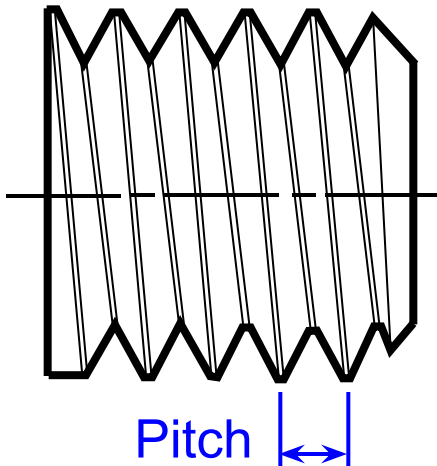
## ***Pitch***

The distance between crests of threads.

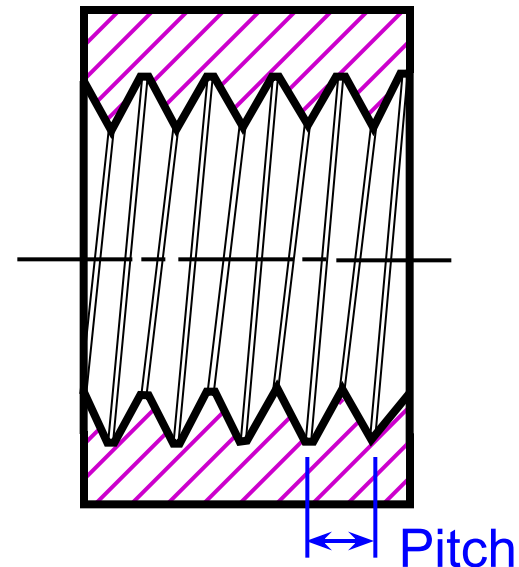
## **Lead**

The distance a screw will advance when turned  $360^\circ$ .

## **External Thread**



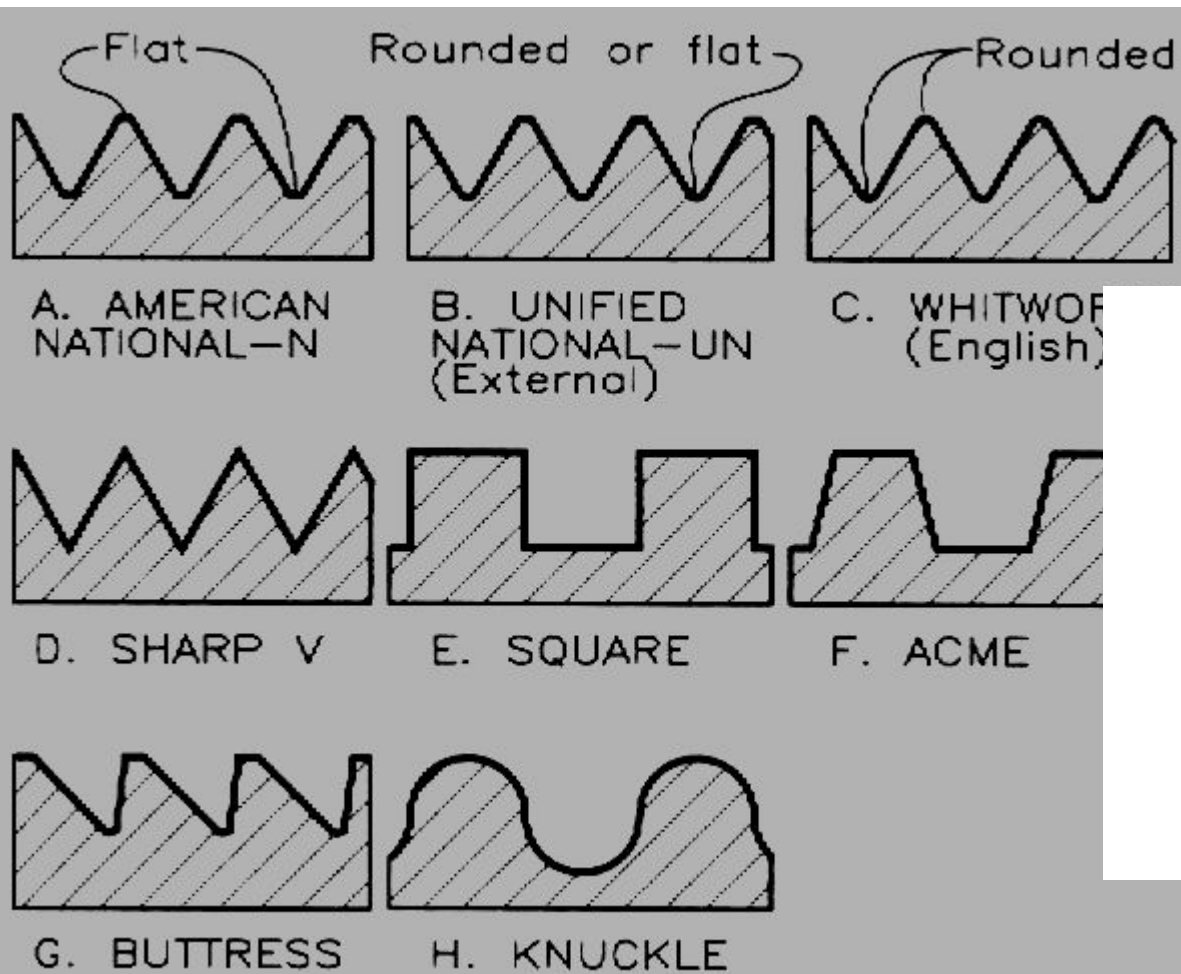
## **Internal Thread**



# THREAD TERMINOLOGY

## Thread Form

Form is the profile shape of the thread.



**Example :**

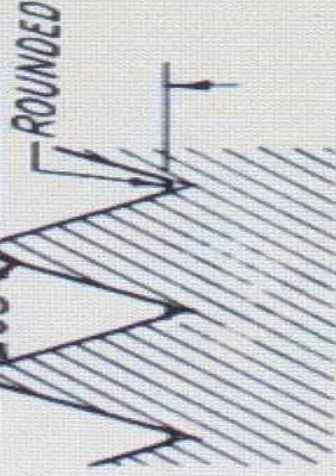
“knuckle thread form”







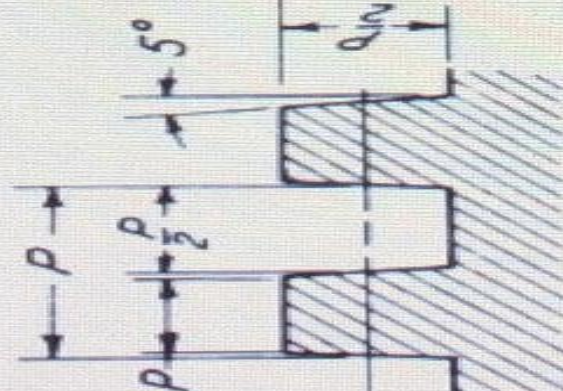
SHARP V



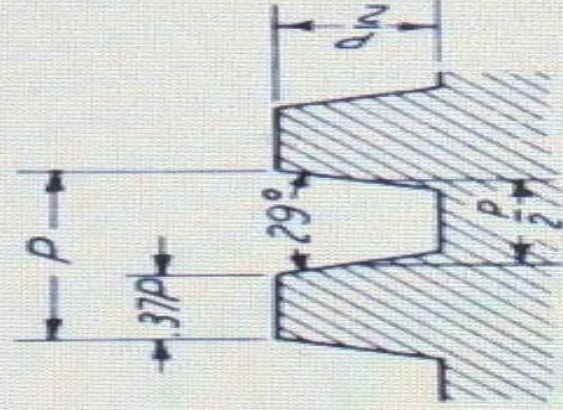
UNIFIED



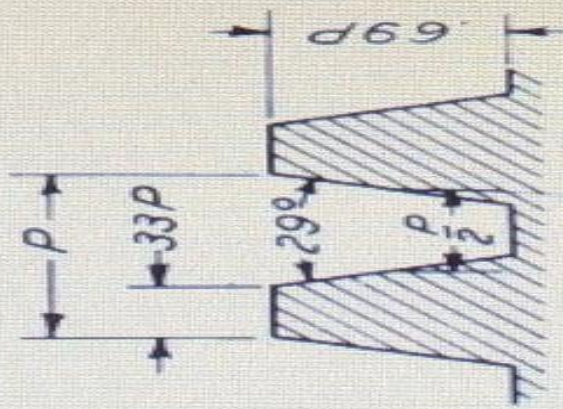
WHITWORTH



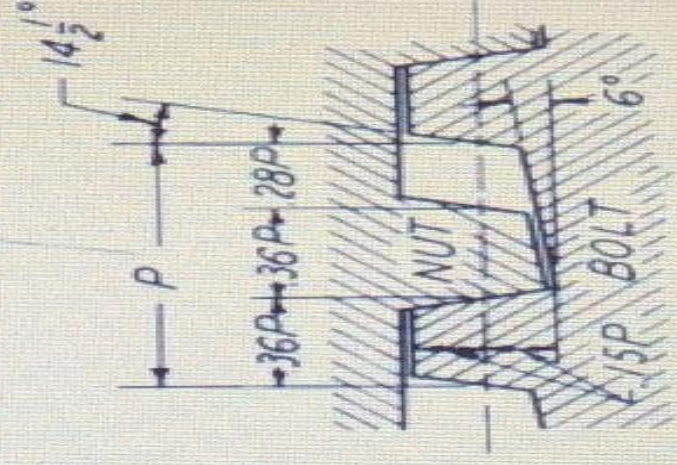
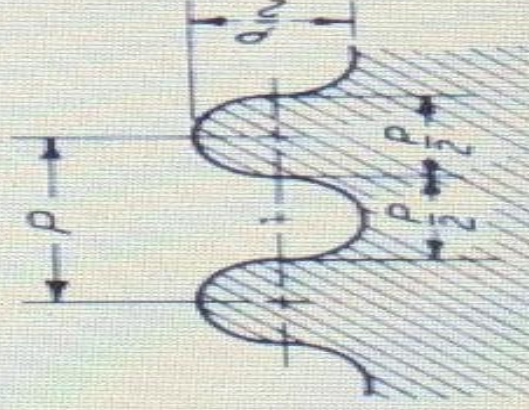
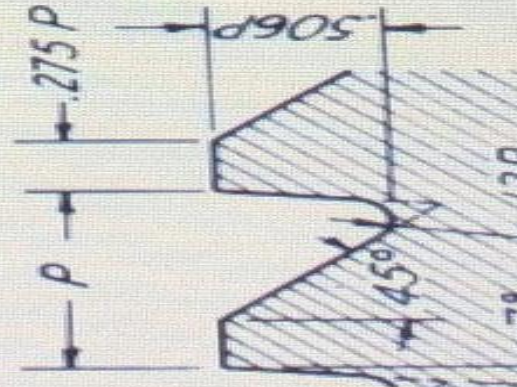
MODIFIED SQUARE  
45° INCLUDED ANGLE



ACME



B&S WORM



NUT  
BOLT  
6°



# EXTERNAL THREAD CUTTING

## Tools

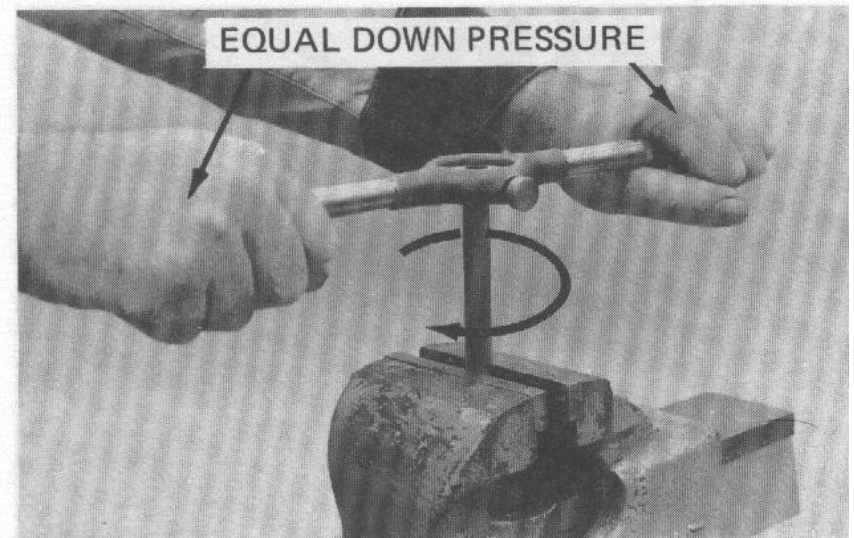
- Threading Die



- Die stock



## Operation





# INTERNAL THREAD CUTTING

## Tools

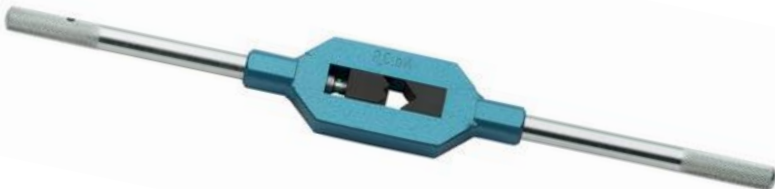
- Twist drill



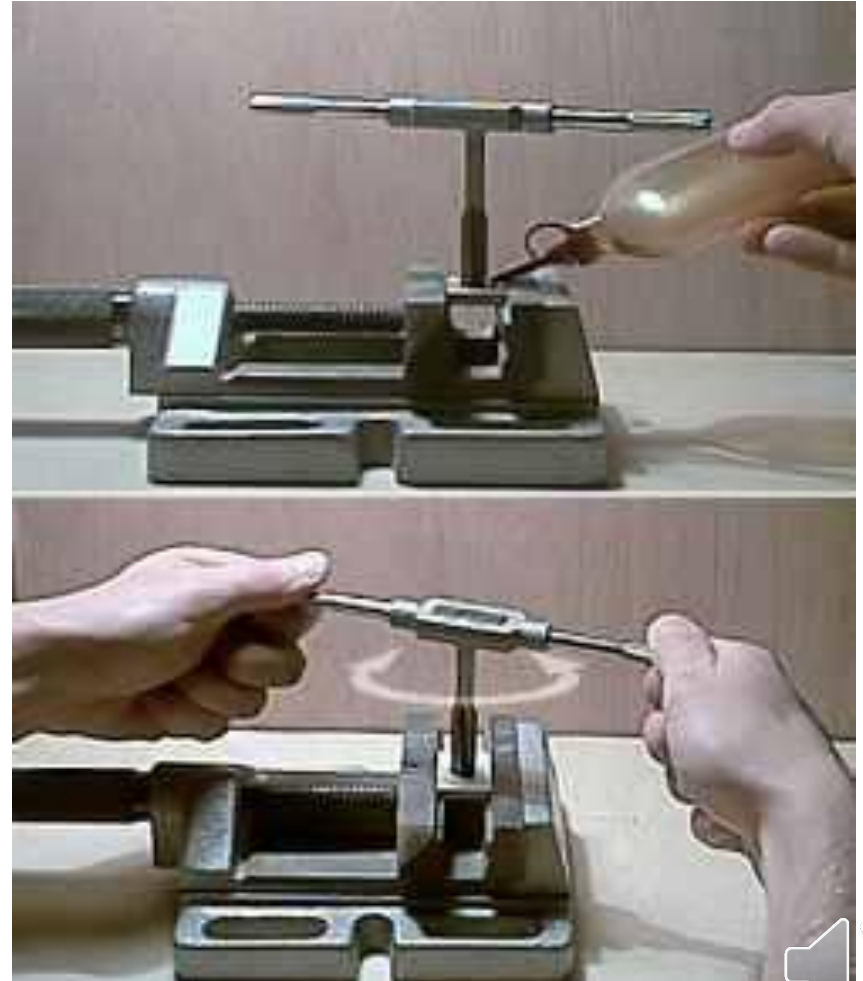
- Tap



- Tap wrench

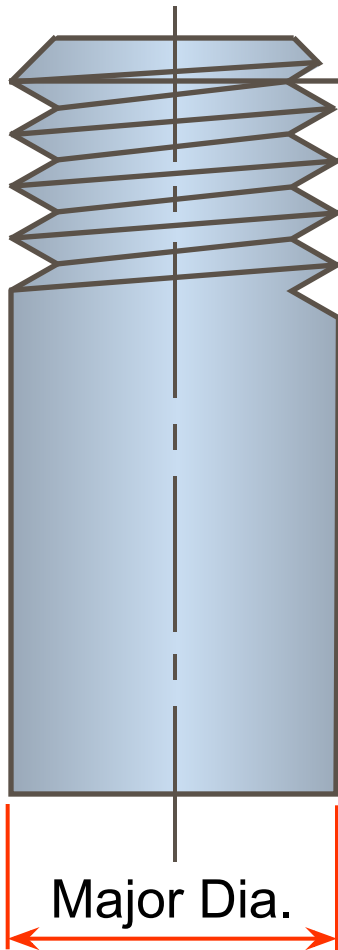


## Operation

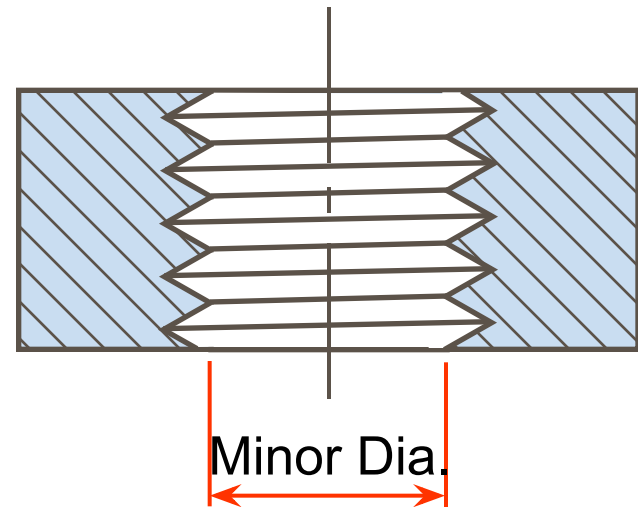


# COMPARISON OF THREAD CUTTING

## External Thread



## Internal Thread





# *Thread Drawing*



# THREAD REPRESENTATION

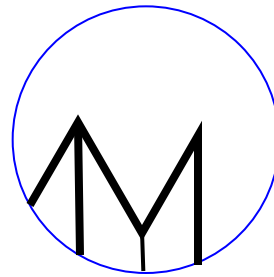
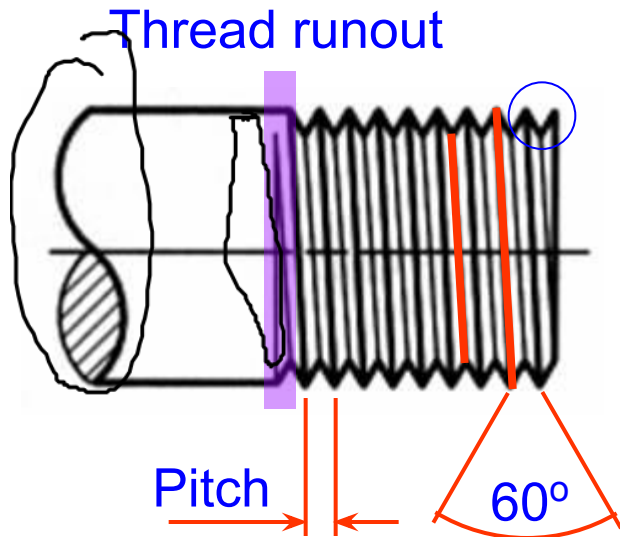
1. ***Detailed*** representation
2. ***Schematic*** representation
3. ***Simplified*** representation



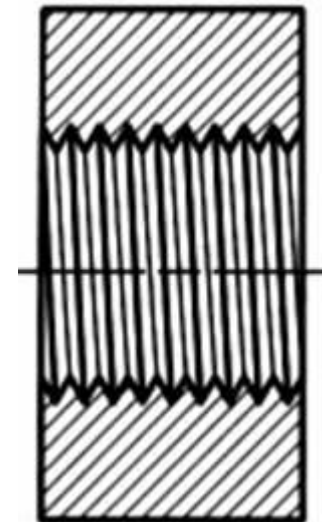
# DETAILED REPRESENTATION

- Use *slanting lines* to represent crest and root.
- Roots and crest are drawn in *sharp* Vs.

## External thread



## Internal thread

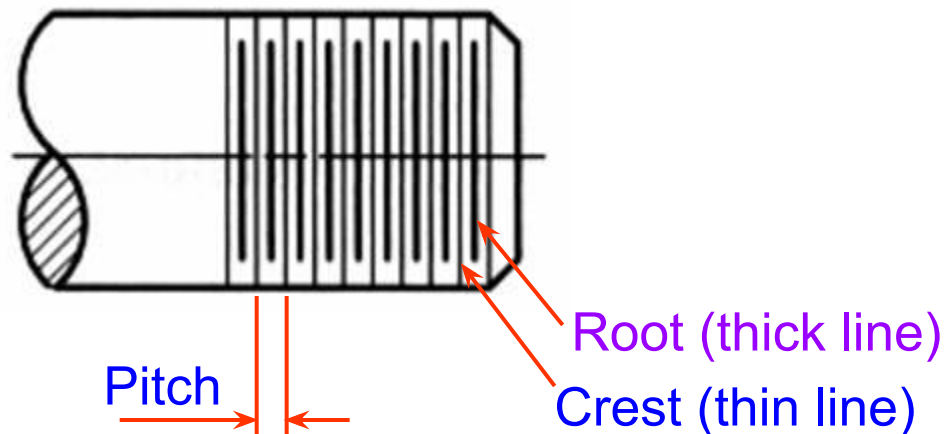




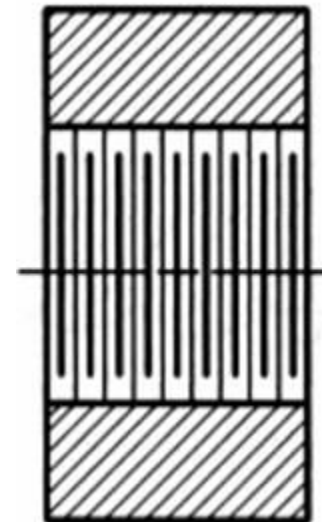
# SCHEMATIC REPRESENTATION

- Use alternate *long* and *short* lines for representing *crests* and *roots* of the thread, respectively.

## External thread



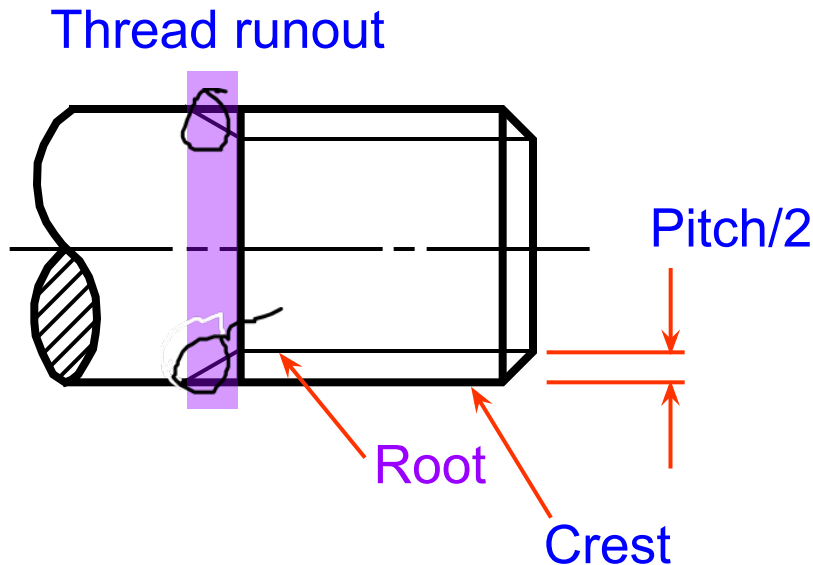
## Internal thread



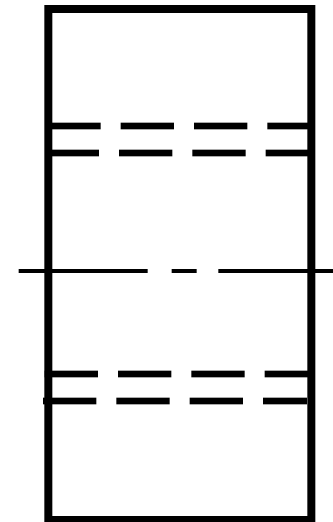
# SIMPLIFIED REPRESENTATION

- Use *thick continuous lines* for representing *crest* and *thin continuous lines* for representing *root* of the thread, respectively.

## External thread



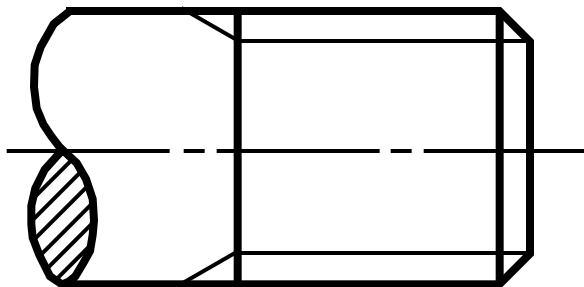
## Internal thread



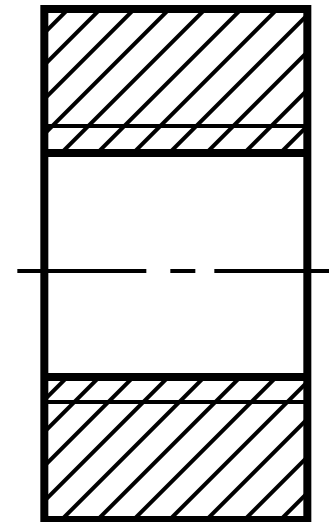
# SIMPLIFIED REPRESENTATION

- Use *thick continuous lines* for representing *crest* and *thin continuous lines* for representing *root* of the thread, respectively.

## External thread



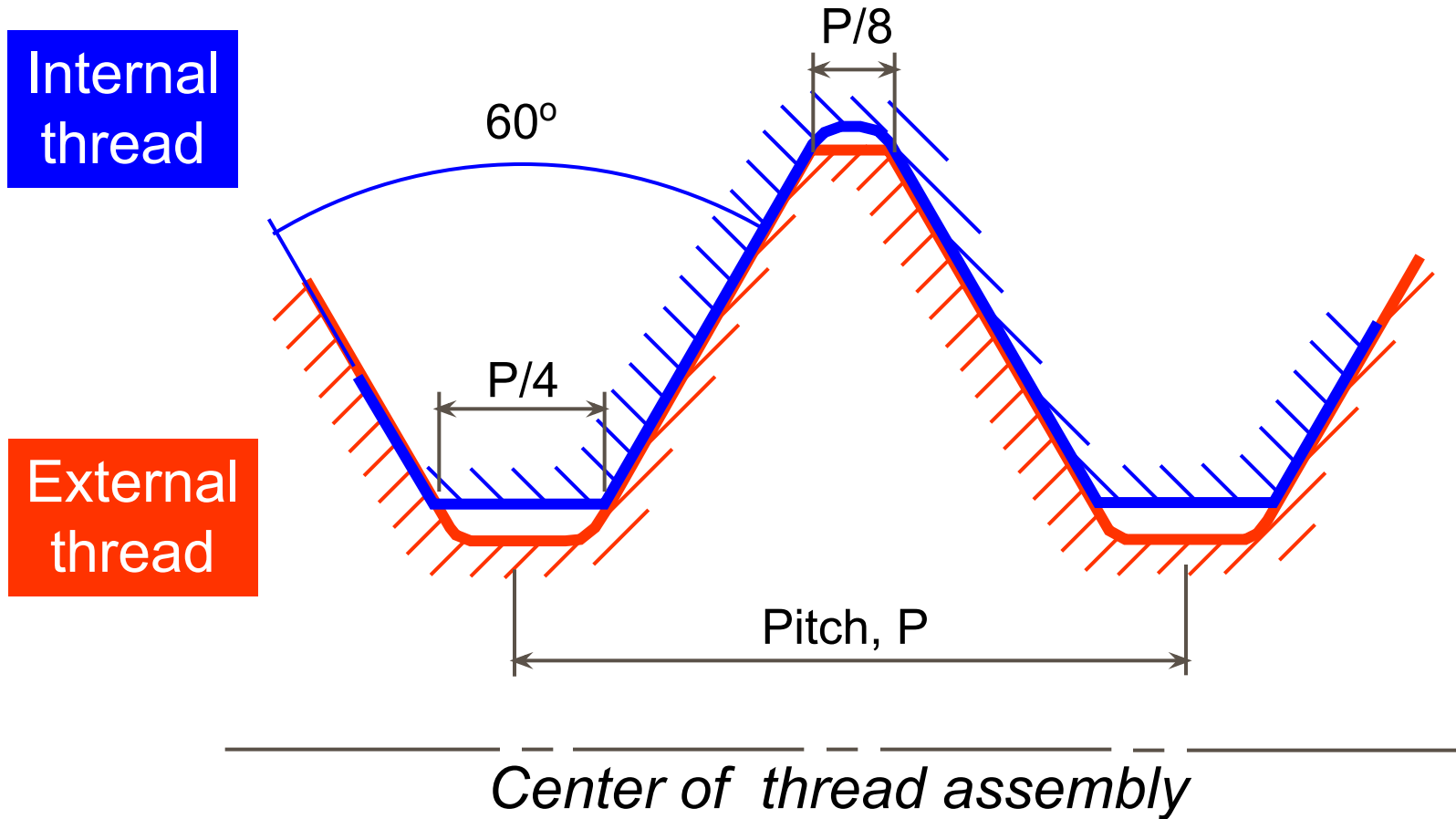
## Internal thread



Sectional view



# ISO (METRIC) THREAD



Thread assembly occurs if and only if both (internal & external) thread have an equal ***nominal size (or diameter)*** and ***pitch***.

# METRIC COARSE THREAD

| Nominal size | Major diameter | Pitch | Minor diameter |
|--------------|----------------|-------|----------------|
| M6           | 6.00           | 1.00  | 4.92           |
| M8           | 8.00           | 1.25  | 6.65           |
| M10          | 10.00          | 1.50  | 8.38           |
| M12          | 12.00          | 1.75  | 10.11          |

Metric thread

In thread *drawing*, the following relationship is used.

$$\text{Minor diameter} = \text{Major diameter} - \text{Pitch}$$



# METRIC FINE THREAD

| Nominal size | Major diameter | Pitch | Minor diameter |
|--------------|----------------|-------|----------------|
| M8           | 8.00           | 0.75  | 7.188          |
|              |                | 1.00  | 6.917          |
| M10          | 10.00          | 0.75  | 9.188          |
|              |                | 1.00  | 8.917          |
|              |                | 1.25  | 8.647          |

In thread *drawing*, the following relationship is used.

$$\textit{Minor diameter} = \textit{Major diameter} - \textit{Pitch}$$





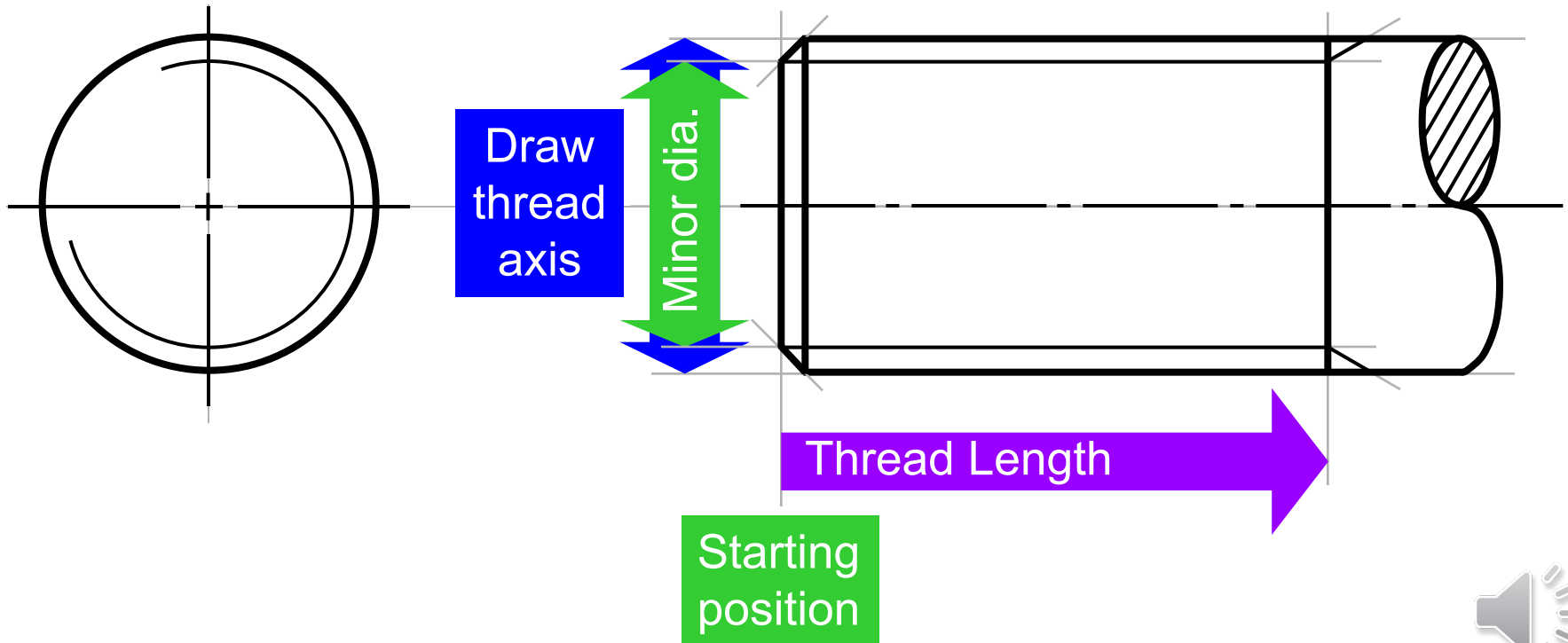
# DRAWING STEPS OF EXTERNAL THREAD

Draw an arc that  
represents a root.

Draw a circle that  
represents a crest.

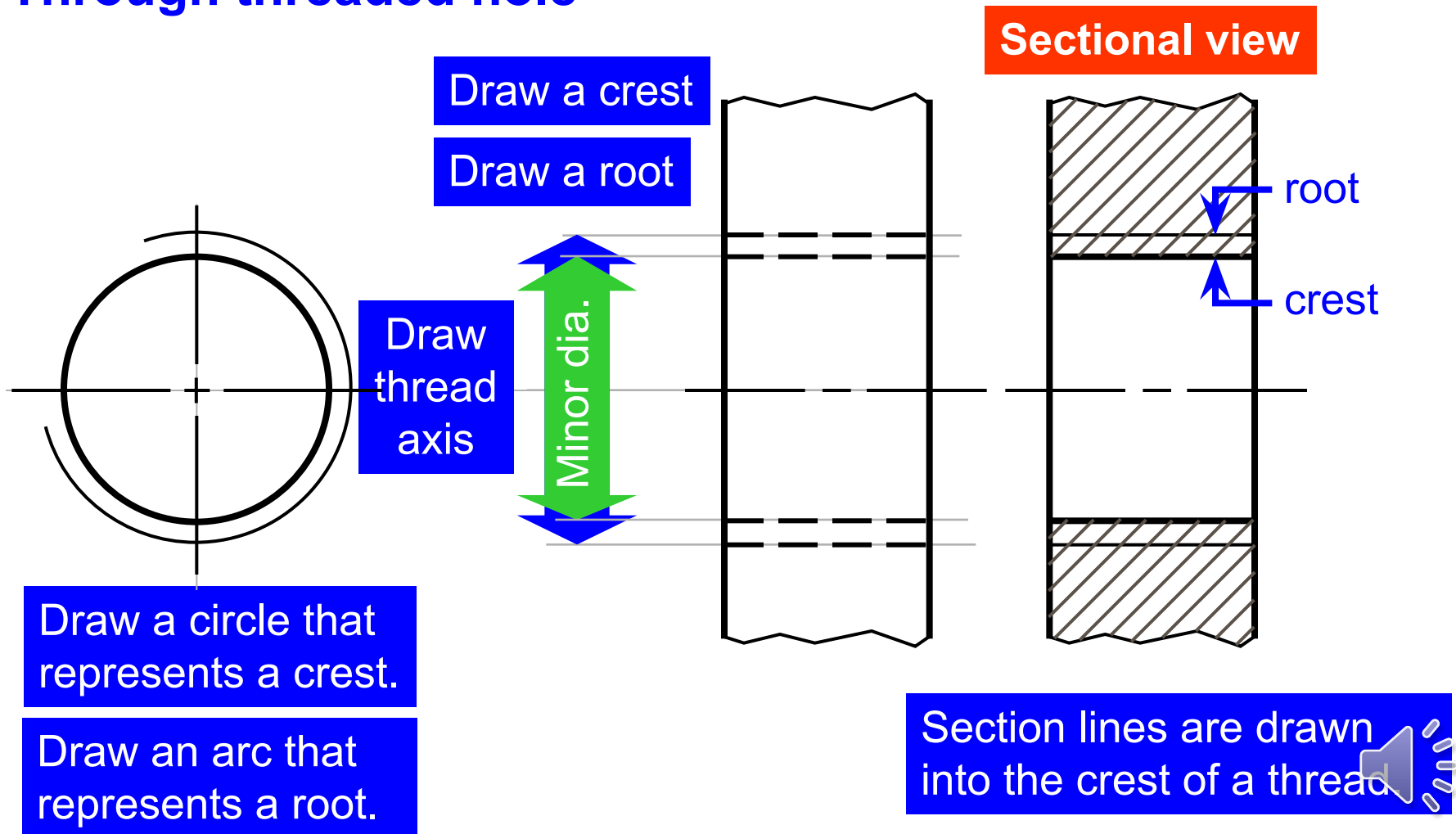
Draw  
45° Chamfer

Draw line making  
30° with thread axis



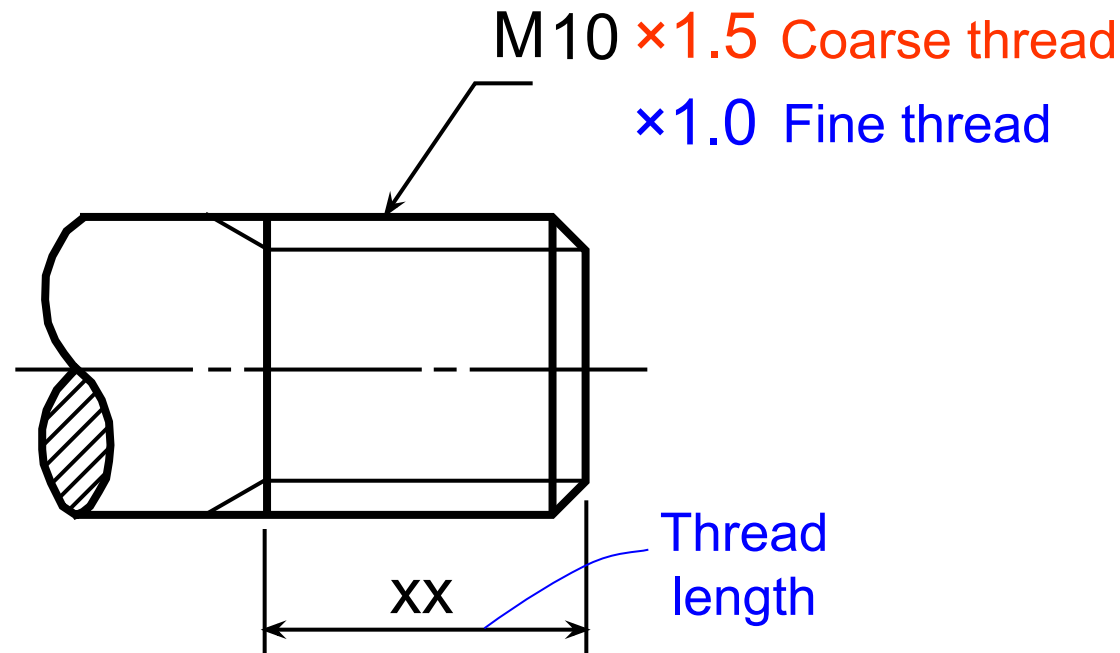
# DRAWING STEPS OF THREADED HOLE

## Through threaded hole

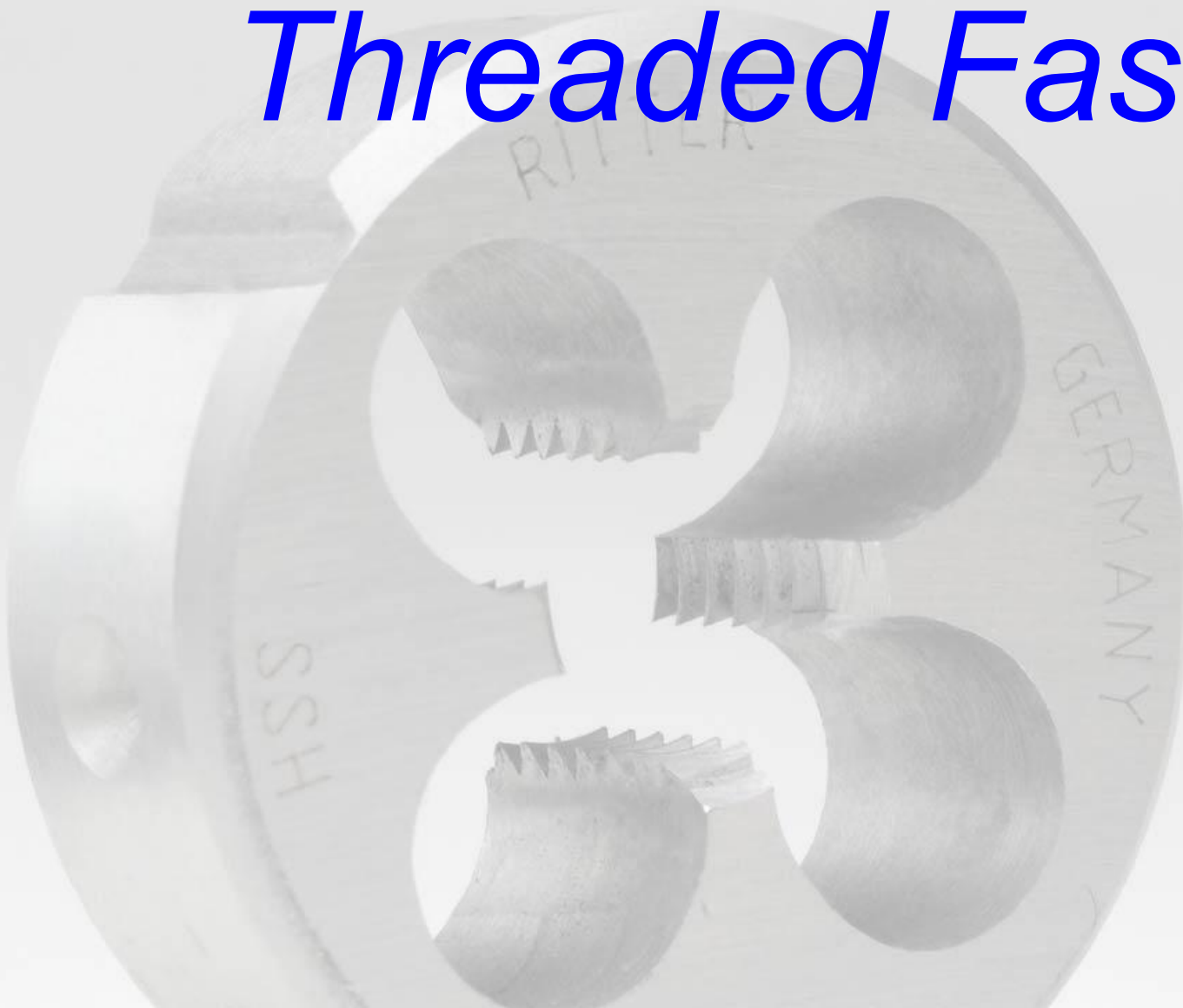


# DIMENSIONING EXTERNAL THREAD

- Use *local note* to specify :- *thread form*, *nominal size*, *pitch* (if it is a fine thread)
- Use *typical method* to specify :- *thread length*.



# *Threaded Fastener*

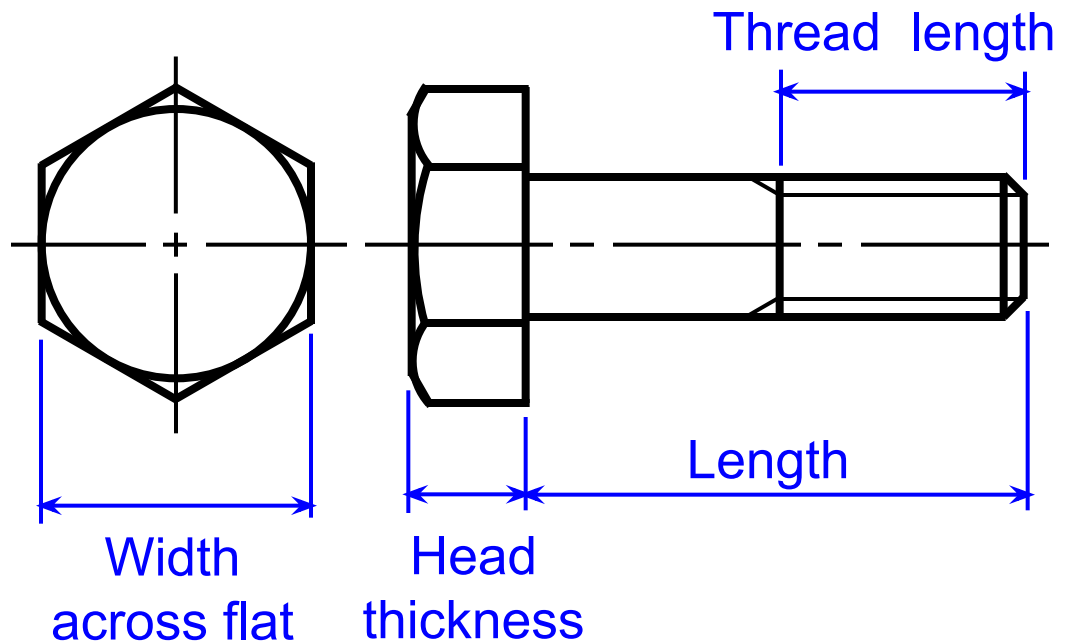


# BOLT : Terminology

***Bolt*** is a threaded cylinder with a head.



Hexagonal head  
bolt and nut

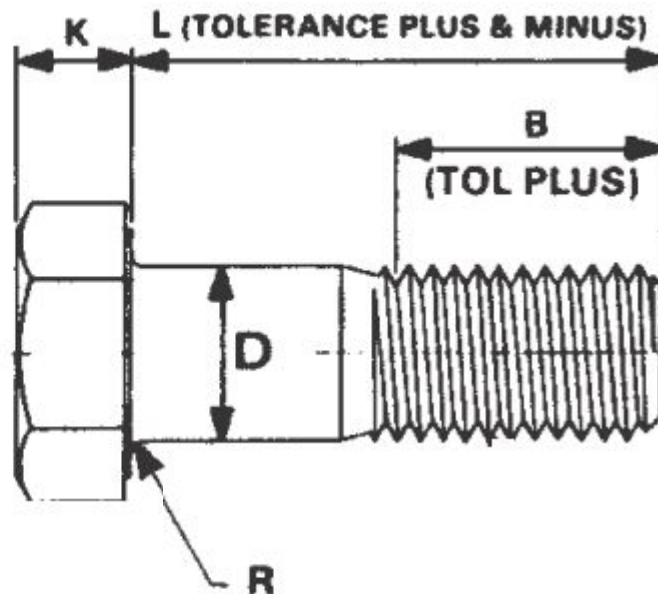
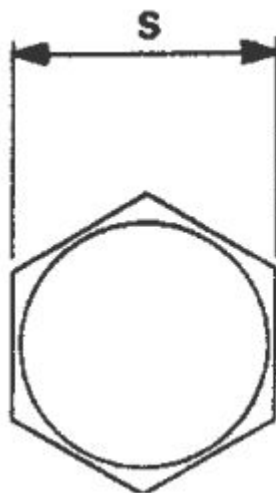


Dimensions of bolt's head are  
listed in table

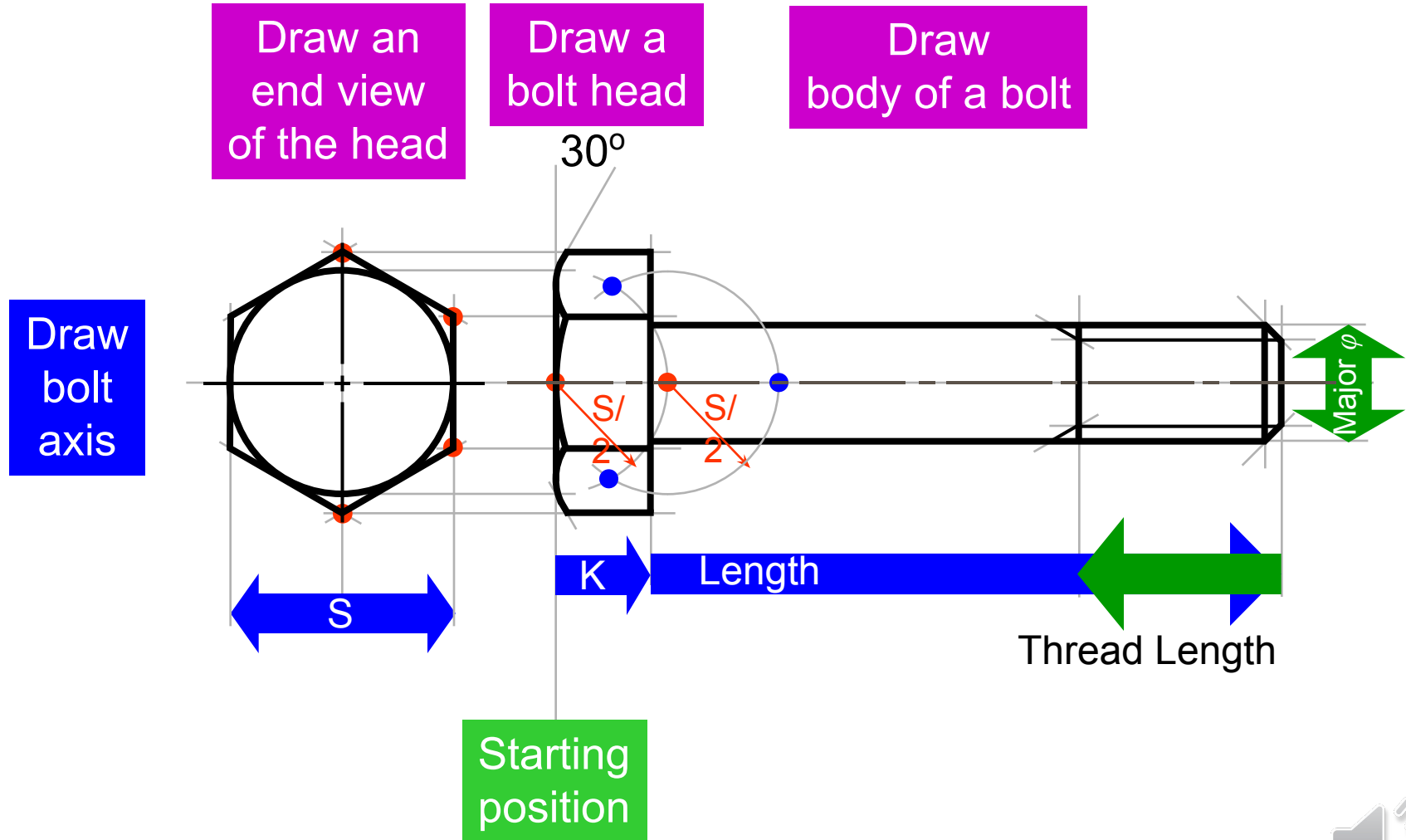


# ANSI/ISO METRIC HEX BOLTS

| Nominal Screw Diameter & Thread Pitch | D             |       | S                  |       | K           |       | R                  | B Thread Length |                   |             |
|---------------------------------------|---------------|-------|--------------------|-------|-------------|-------|--------------------|-----------------|-------------------|-------------|
|                                       | Body Diameter |       | Width Across Flats |       | Head Height |       | Fillet Radius Min. | For L ≤ 125     | For L > 125 ≤ 200 | For L > 200 |
|                                       | Min.          | Max.  | Min.               | Max.  | Min.        | Max.  |                    |                 |                   |             |
| M6 x 1                                | 5.82          | 6.00  | 9.78               | 10.00 | 3.85        | 4.15  | 0.3                | 18              | 24                | 37          |
| M8 x 1.25                             | 7.78          | 8.00  | 12.73              | 13.00 | 5.10        | 5.50  | 0.4                | 22              | 28                | 41          |
| M10 x 1.5                             | 9.78          | 10.00 | 15.73              | 16.00 | 6.17        | 6.63  | 0.4                | 26              | 32                | 45          |
| M12 x 1.75                            | 11.73         | 12.00 | 17.73              | 18.00 | 7.24        | 7.76  | 0.6                | 30              | 36                | 49          |
| M14 x 2                               | 13.73         | 14.00 | 20.67              | 21.00 | 8.51        | 9.09  | 0.6                | 34              | 40                | 53          |
| M16 x 2                               | 15.73         | 16.00 | 23.67              | 24.00 | 9.68        | 10.32 | 0.6                | 38              | 44                | 57          |
| M20 x 2.5                             | 19.67         | 20.00 | 29.16              | 30.00 | 12.12       | 12.88 | 0.8                | 46              | 52                | 65          |
| M24 x 3                               | 23.67         | 24.00 | 35.00              | 36.00 | 14.56       | 15.44 | 0.8                | 54              | 60                | 73          |
| M30 x 3.5                             | 29.67         | 30.00 | 45.00              | 46.00 | 17.92       | 19.48 | 1.0                | 66              | 72                | 85          |



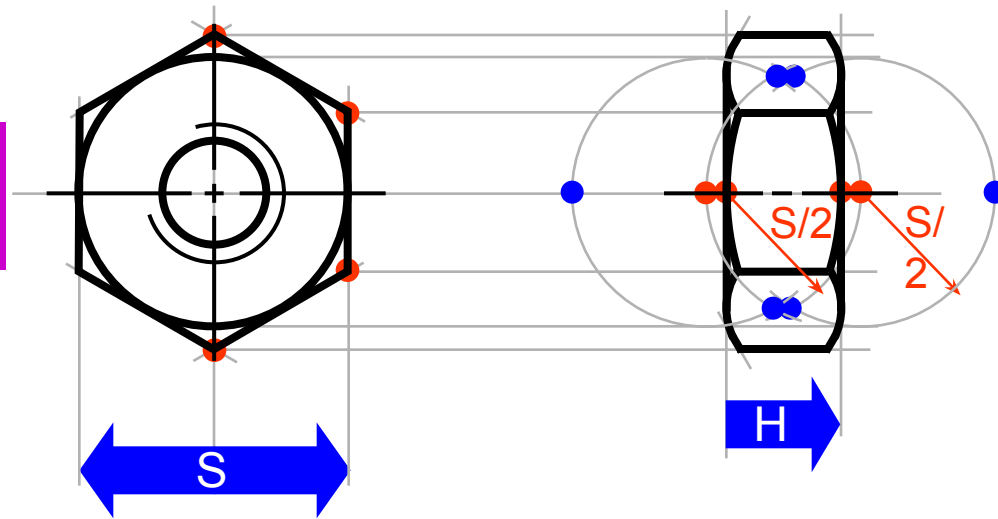
# BOLT : Drawing steps





# NUT : Drawing steps

Draw an end view of the nut



Dash lines represent a threaded hole are omitted for clarity.

